

## PROFESSIONAL SUMMARY

- Optical & Photonics R&D Engineer** specializing in the end-to-end development of metasurface-enabled nonlinear optics and infrared systems. Proven track record in translating complex physics models into high-performance hardware, including a LiNbO<sub>3</sub>-based source achieving record-level efficiency. Expert in FDTD (CST, HFSS) workflows, with a history of securing non-dilutive funding (e.g., \$300k NSF seed grant) and automating high-throughput characterization pipelines to accelerate product development cycles.

## SKILLS

- Design & Simulation:** CST Microwave Studio (FDTD), Ansys HFSS, Zemax OpticStudio, SolidWorks, electromagnetic modeling, diffractive optics, nonlinear optics
- Experimental Hardware:** IR & ultrafast lasers (Ti:Sapphire, CO<sub>2</sub>, Diode), OPA, EOM, Oscilloscope, Spectrum Analyzer, Autocorrelator, Lock-in Amplifier, DAQ systems
- Lab Techniques:** Precision Optical Alignment, Cryogenic Metrology, Heterodyne Detection, Noise Floor Characterization, Stability Analysis
- Programming & Analysis:** Python, MATLAB (DSP, FFT), LabVIEW, Automated Characterization Pipelines

## EXPERIENCE

Doctoral researcher | UC San Diego, La Jolla CA, USA

Aug 2020 – Present

- EM Simulation & Design (FDTD):** Validated tunable infrared (800 nm – 300  $\mu$ m) electro-optical systems using **CST Microwave Studio** and **HFSS**. Optimized nonlinear conversion efficiency and spectral tunability through iterative design–fabrication–measurement cycles
- Nonlinear Optical Engineering:** Engineered a **LiNbO<sub>3</sub>-based** nonlinear light source with diffractive optics, achieving record-high **0.13% far-IR (THz) generation**
- Cryogenic Metrology & Alignment:** Integrated electro-optical assemblies into advanced optical characterization, enabling direct field measurements at **cryogenic temperatures** with **sub-micron alignment stability**
- Nanophotonic Device Optimization:** Designed and characterized a gold-on-silicon nanophotonic array for extreme nonlinear coupling with **120x field enhancement**
- Data Automation & High-Throughput:** Developed automated **MATLAB and Python pipelines** for high-throughput optical material characterization, **reducing data processing time by over 50%**

Research Trainee | Materials Research Science &amp; Engineering Centre (MRSEC) UC San Diego, Apr 2023 – Mar 2024

- Developed infrared metrology schemes for plasmonic quantum-material systems, demonstrating frequency-tunable terahertz sensing using self-assembled Ag nanoparticles on Si substrates
- Contributed to **NSF proposal leading to \$300k seed funding**.

Graduate Research associate | Molecular Dynamics Lab - IISER Kolkata, West Bengal, India

Apr 2018 – May 2019

- Demonstrated time-of-flight (TOF) measurements of oxygen clusters using high-speed electronics and electro-optics
- Developed Python codes for using large-scale TOF data to obtain molecular velocity distribution and auto-identify clusters

Visiting research scholar | University of Duisburg-Essen, Duisburg, Germany

May 2017 – Aug 2017

- Implemented sub-picosecond ultrafast infrared pulse characterization using Frequency Resolved Optical Gating (FROG)
- Received scholarship of €2000 through DFG(German Research Foundation) to spend a summer doing research in Germany

Technical Lead &amp; Associate Instructor | UC San Diego, La Jolla CA, USA

(Concurrent with PhD Research)

- Directed lab operations and safety protocols for physics labs, managing a team of 50+ PhD-level assistants over 5 years
- Streamlined transition of physics concepts into lab modules, showing communication skills certified by the Fleet Science Center

## EDUCATION

PhD in Experimental Physics (Applied Optics &amp; Quantum Materials), University of California San Diego

April 2026

BS &amp; MS in Physics, Indian Institute of Science, Education &amp; Research, Kolkata ( BS-MS CGPA 9.0/10)

June 2019

## PROJECTS BEYOND DOCTORAL RESEARCH

- High-Precision Infrared Michelson Spectrometer** | Skills: Precision Alignment, Interferometry
  - Engineered a portable, wide-band (600 nm – 20  $\mu$ m) spectrometer featuring sub-micron interferometric stability control
  - Automated real-time signal processing in MATLAB, reducing spectral acquisition time and enabling on-the-fly characterization in field environments

## 2. Tunable Multi-THz Nanophotonic Filters | Skills: CST Microwave Studio, FDTD

- Optimized gold nano-slit geometries using CST Microwave Studio to achieve high-Q spectral selectivity for multi-THz applications
- Designed for mechanical flexibility, ensuring high optical performance and resonance stability when integrated into flexible substrates

## 3. Automated Phase-Retrieval & Error Correction Pipeline | Skills: Python, Kramers-Kronig

- Developed a robust Python/MATLAB pipeline utilizing Kramers–Kronig relations to automatically correct phase errors caused by system misalignment
- Improved measurement accuracy for complex reflectance data, enabling high-fidelity characterization of quantum materials and thin films

## PUBLICATIONS

- A Verma, et al. "Picosecond volume expansion drives a later-time insulator–metal transition in a nano-textured Mott insulator." **Nature Physics (2024)**
- Y Huang, et al. "All-silicon active bound states in the continuum terahertz metamaterials." **Optics & Laser Technology (2024)**
- Peer-Reviewed Publications (2), Preprints (1), Manuscripts in Preparation (3)

## AWARDS AND RECOGNITION

- INSPIRE Fellowship (Top 1% nationwide, India)
- UC San Diego Physics Excellence Award (\$11,000)
- UC San Diego Physics Chair's challenge award (2024)
- Shortlisted for Best MS Thesis in Physics (2019).

## PROFESSIONAL ACTIVITIES

- Peer-reviewed scientific articles for Nature Physics, Proceedings of the National Academy of Sciences (PNAS), Nanoletters and Phys. Review B
- Certified by Fleet Science Center for science communication (2023) Balboa Park, San Diego CA